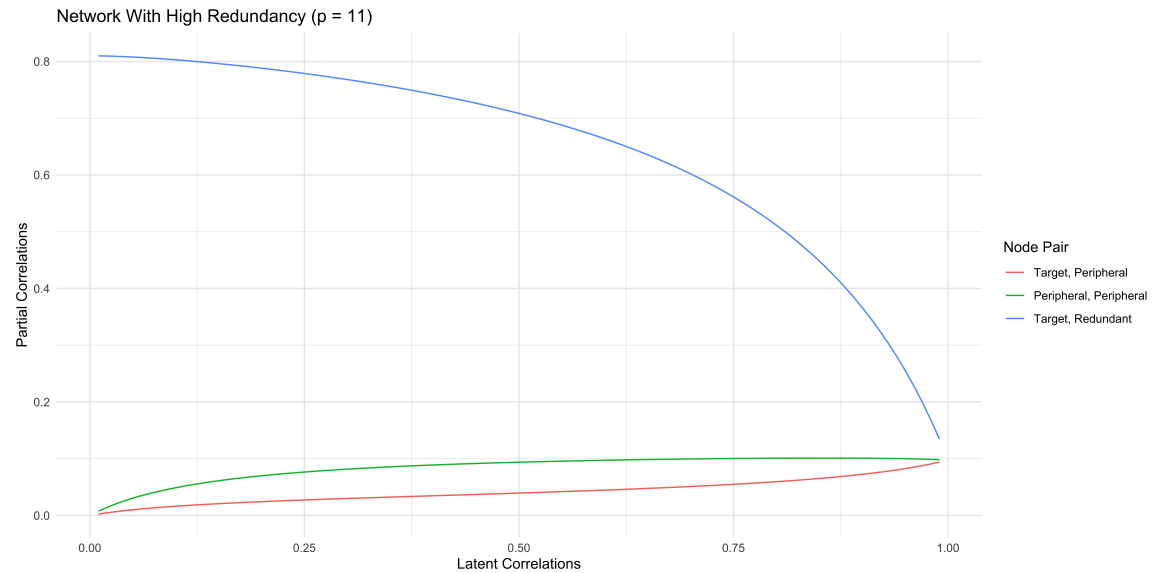


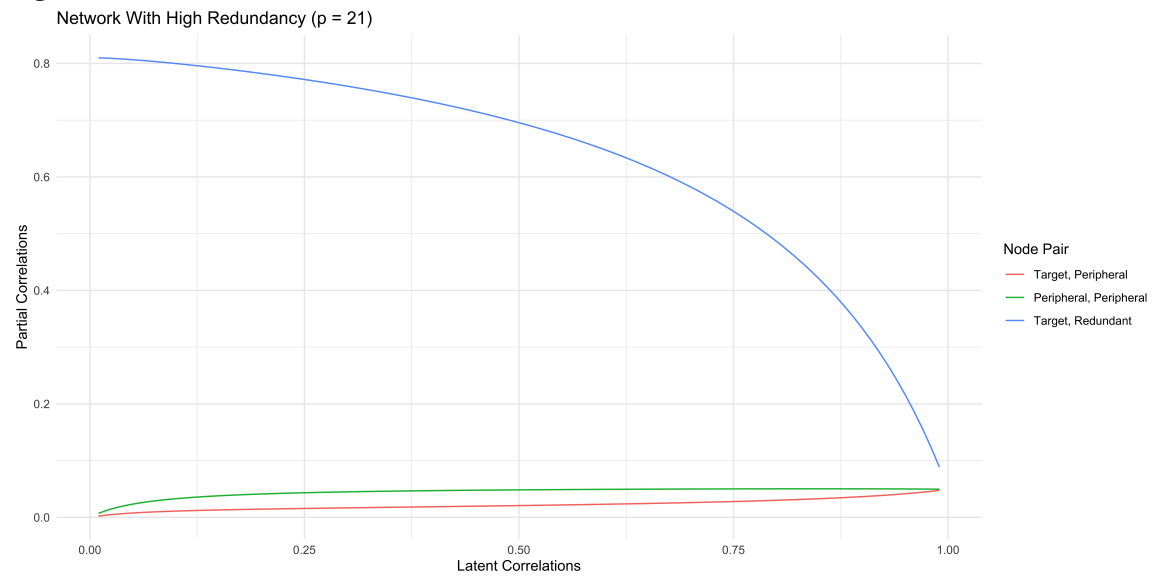
Redundancy & Latent Correlations

In our design, each item had unit variance, and the correlation between the redundant items was equal to $\lambda_i \lambda_j$, the product of their loadings. Likewise, the correlation between any two nodes that were not the redundant node pair was given by $\lambda_i \lambda_j \rho(\psi_i, \psi_j)$, the product of their loadings and the correlation between their respective factors. Furthermore, transforming Σ into a partial correlation matrix meant obtaining the partial correlation between any two nodes after removing the variance explained by all other nodes in the network. In a separate simulation, we systematically varied the degree of redundancy, over a sequence of latent correlations from 0.01 to 0.99 by increments of 0.01. In each iteration, the partial correlation between three node pairs was saved (i.e., the target and a peripheral node, a peripheral and another random peripheral node, and the target and redundant node). The loadings of every item were fixed at 0.9. Figure 1 and 2 show the trend of each node pair for each level of p used in our main simulation. In both plots, as the latent correlations increased, the partial correlation between the redundant node pair decreased. Similarly, the distance between the other two node pairs decreased. At a correlation of approximately 0.7, the peripheral node pair plateaued, while the target-peripheral pair's partial correlation rose quickly to close the remaining gap between the two. This was also the point at which the redundant nodes' partial correlations sharply decreased and became less impactful. If the correlation between the redundant items in Σ depends on $\lambda_i \lambda_j$, while the correlation between all other item pairs depends on $\lambda_i \lambda_j \rho(\psi_i, \psi_j)$, then as the latent correlations increase, the correlation between all item pairs except the target-redundant pair, should increase. As the peripheral items start to share more and more information with other variables, their nodes start to compete with each other for variance at the partial correlation level. As a result, the redundant node becomes less impactful, because more of its variance can be explained by other nodes in the network. Correspondingly, as the redundant node starts to become just another node in the network, the target and redundant node partial correlation increases and eventually converges with the peripheral node pair.

In both expected influence and strength, the measures were computed by taking the sum of a node's connections (the absolute weights in strength). In all iterations, the target node's connections with all other nodes beside the redundant node significantly decreased. The peripheral nodes connections with each other consistently stayed the same with minor fluctuations. Therefore, when comparing the target node to its counterpart in the true node set, there was a significant difference when the redundant node was removed.

Figure 1

Note. The impact of latent correlations on partial correlations in a network with a redundant node. Lines denote different node pairings. The x-axis denotes latent correlations from 0.01 to 0.99 by increments of 0.01. The y-axis denotes the partial correlations of each node pair.

Figure 2

Note. The impact of latent correlations on partial correlations in a network with a redundant node. Lines denote different node pairings. The x-axis denotes latent correlations from 0.01 to 0.99 by increments of 0.01. The y-axis denotes the partial correlations of each node pair.